GNCORE / LOCAL REVIEW / 7.0.0rc1

Skill improvement review

All 30 developer domains were revised and the local package checks passed. A fair AI-quality comparison remains incomplete after an account usage limit. This report separates verified engineering work from unproven performance claims.

GNCORE / INTERNAL ENGINEERING VALIDATION

Evidence progress, not product performance

Actual local checks | Each row has its own units | No overall quality percentage



* The fair AI comparison stopped at the account usage limit. A >95% all-domain quality result is not established.

Earlier shared-context answer scores are development evidence only; they are withdrawn from quality comparisons.

The 160 code fixtures test inspected development artifacts; they do not prove which guide caused the result. Narrow UI checks remain open.

COMPLETE DOMAIN TABLE

Verified build status; quality target still open

Every category passed the applicable catalog/merge checks and installation verification. These check files and structure, not AI effectiveness. No domain currently has a completed fair >95% comparative quality measurement in this cycle.

Domain	Workflows	Bundle checks	Additional executed task evidence*	AI quality >95%
AI Engineering	6	Passed	No domain runtime measurement	Unproven
Automation	2	Passed	No domain runtime measurement	Unproven
Cloud DevOps SRE	15	Passed	No domain runtime measurement	Unproven
Code Build	85	Passed	19/19 allocation fixtures	Unproven
Computer Vision	5	Passed	No domain runtime measurement	Unproven
Cybersecurity	12	Passed	No domain runtime measurement	Unproven
Data Engineering	4	Passed	No domain runtime measurement	Unproven
Data Science	18	Passed	No domain runtime measurement	Unproven
Databases	5	Passed	No domain runtime measurement	Unproven
Deep Learning	9	Passed	No domain runtime measurement	Unproven
Design UX	12	Passed	No domain runtime measurement	Unproven
Developer Research	4	Passed	No domain runtime measurement	Unproven
Developer Tools	2	Passed	No domain runtime measurement	Unproven
Distributed Systems	4	Passed	No domain runtime measurement	Unproven
Documentation	3	Passed	No domain runtime measurement	Unproven
Embedded Systems	4	Passed	No domain runtime measurement	Unproven
Game Development	1	Passed	No domain runtime measurement	Unproven
Machine Learning	21	Passed	No domain runtime measurement	Unproven
MLOps	10	Passed	No domain runtime measurement	Unproven
Mobile Development	1	Passed	No domain runtime measurement	Unproven
Natural Language Processing	5	Passed	No domain runtime measurement	Unproven
Neural Networks	6	Passed	14/14 math fixtures	Unproven
Observability	7	Passed	No domain runtime measurement	Unproven
Performance Engineering	5	Passed	No domain runtime measurement	Unproven
Python	3	Passed	No domain runtime measurement	Unproven
Release Engineering	2	Passed	No domain runtime measurement	Unproven
Systems Programming	4	Passed	No domain runtime measurement	Unproven
Testing	19	Passed	7/7 checker fixtures	Unproven
Web Backend	60	Passed	No domain runtime measurement	Unproven
Web Frontend	40	Passed	5 UI checks passed; 1 open	Unproven

* Executed examples are synthetic development artifacts, not full real-project evaluations. Each of four development conditions passed the listed code fixtures and five browser checks; four narrow-layout checks remain unmeasured. The fair fresh-context comparison is incomplete.

WHAT / WHY / WHERE

A lighter entrypoint, with stronger completion rules

Each domain now has a short task-selection guide, category-specific failure boundaries and a map into a hashed WORKFLOWS.md resource. Main entrypoints decreased from 309,898 to 175,877 bytes (43.2% smaller). This measures file bytes only; conditional resources still consume context when read.

Change	Reason	Location
Completion boundaries in all 30 domains	Infer essential missing requirements while preserving the requested stack, public API and scope.	catalog/category-completion-checks.json
Conditional procedures in all 30 domains	Read the matching workflow and necessary references instead of every procedure by default.	scripts/build_catalog.py; each category/WORKFLOWS.md
Ten procedure refinements; eight discovery summaries	Correct actual scope/correctness problems and contradictory discovery text; retain before/after lineage.	workflow-quality-changes.json; migration quality_revisions
Three focused completion refinements	Explicit runtime secrets; a runnable, honest documentation example; actual native ABI boundary rules.	focused-guidance-changes.json; focused-review-notes.json

Concrete procedure corrections

Area	Correction
Frontend / design	Remove a fixed organic theme; route actual workflow sections; require designer handoff only when applicable.
Automation / databases	Honor existing authorization; coordinate old/new writers and resumable reconciliation before contraction.
Python / releases	Build the wheel from its sdist; promote the same verified bytes once per intended target.
Spatial / neural / native work	Allow suitable geodesic methods; freeze stochastic state for relevant gradient checks; separate checkable pointer bounds from caller provenance obligations.

The merge check retained all 374 workflow IDs: 364 procedures are unchanged and 10 have reviewed revisions. Existing resource declarations and source licenses are preserved. Long files alone are not treated as evidence of quality.

BUILD / TEST / INSTALL

What was actually executed

Check	Observed result	Practical limit
Final full Python suite	151 passed; 1 skipped	Local Windows/Python 3.14. Symlink creation was unavailable; this is not a full platform/interpreter matrix.
Catalog and merge integrity	30 categories; 374 workflows; 1,660 resources verified; 73 helpers parsed	Hash/link/procedure and syntax checks; upstream helpers were not executed.
Generated allocation code	16/16 frozen cases + 3/3 exploratory cases, in all four development conditions	Pure synthetic integer allocation; no general code-quality percentage.
Generated neural-network math	14/14 cases, in all four conditions	CPU Python loss/gradient fixtures including central differences; no GPU training result.
Generated contract checkers	7/7 fixtures, in all four conditions	Correct implementation plus six seeded mutants; not real-project bug-finding recall.
Generated offline web pages	5/5 real browser checks in all four conditions	Blank/invalid/valid gating, selected confirmation and keyboard focus. 390px could not be established.
Focused documentation example	Copied offline example executed successfully	Acceptance/pending/duplicate/completed assertions. No real HTTP contract verification.
Built and installed distributions	Wheel built from sdist; Twine checks passed; 1,690 installed files verified	Isolated installed import, source/resource byte comparison, directory export and portable ZIP; no upload.

Generated-code tests and browser observations are real, but the development agents were not uniformly isolated from prior chat context. These artifact results do not establish a causal skill benefit. Security findings were static development review only; broad vulnerability detection and production security were not measured.

The first oracle rejected valid integer subclasses, and the inspector initially rejected benign error formatting. Both harness problems were corrected uniformly; original logs remain available. Neither correction is counted as a GNCore skill gain.

RESEARCH SOURCES / COMPARISON PENDING

Thirty matched guides from fourteen repositories

Pinned source entrypoints were downloaded again and checked against the recorded snapshots after CRLF/LF normalization. This is task-matched comparison, not a ranking of companies, Codex, Plotly, or every available skill. Reference helpers were not executed. Click a source to inspect its exact revision.

GNCore domain	Matched repository / entrypoint	Pinned revision
AI Engineering	anthropics/skills / skills/mcp-builder	ca45fc496a34
Automation	temporalio/skill-temporal-developer / root	5f5a94d448d5
Cloud DevOps SRE	google/skills / skills/cloud/gke-reliability	81b0e8f8f405
Code Build	obra/superpowers / skills/systematic-debugging	2ce8aa15d05b
Computer Vision	huggingface/skills / skills/huggingface-vision-trainer	fabc634a893f
Cybersecurity	affaan-m/ecc / skills/security-review	25278e7d7b6c
Data Engineering	K-Dense-AI/scientific-agent-skills / skills/polars	5f2bc0341354
Data Science	K-Dense-AI/scientific-agent-skills / skills/statistical-analysis	5f2bc0341354
Databases	supabase/agent-skills / skills/supabase-postgres-best-practices	c9be0e931b79
Deep Learning	K-Dense-AI/scientific-agent-skills / skills/pytorch-lightning	5f2bc0341354
Design UX	microsoft/skills / .github/skills/frontend-design-review	00d66a64919e
Developer Research	K-Dense-AI/scientific-agent-skills / skills/scientific-critical-thinking	5f2bc0341354
Developer Tools	affaan-m/ecc / skills/git-workflow	25278e7d7b6c
Distributed Systems	cloudflare/skills / skills/durable-objects	49a31e0b3229
Documentation	anthropics/skills / skills/doc-coauthoring	ca45fc496a34
Embedded Systems	beriberikix/zephyr-agent-skills / skills/zephyr-foundations	ed63cdfb8cdf
Game Development	minhhoit/gamedev-agent-skills / skills/godot/godot-2d-movement	b105e1cf617a
Machine Learning	K-Dense-AI/scientific-agent-skills / skills/scikit-learn	5f2bc0341354
MLOps	huggingface/skills / skills/hf-cloud-sagemaker-production-defaults	fabc634a893f
Mobile Development	vercel-labs/agent-skills / skills/react-native-skills	1a2c79d4e6c4
Natural Language Processing	K-Dense-AI/scientific-agent-skills / skills/transformers	5f2bc0341354
Neural Networks	K-Dense-AI/scientific-agent-skills / skills/pytorch-lightning	5f2bc0341354
Observability	microsoft/skills / .github/plugins/azure-sdk-python/skills/azure-monitor-opentelemetry-py	00d66a64919e
Performance Engineering	affaan-m/ecc / skills/benchmark	25278e7d7b6c
Python	trailofbits/skills / plugins/modern-python/skills/modern-python	2be6433b39e0
Release Engineering	obra/superpowers / skills/finishing-a-development-branch	2ce8aa15d05b
Systems Programming	trailofbits/skills / plugins/rust-review/skills/rust-review	2be6433b39e0
Testing	obra/superpowers / skills/test-driven-development	2ce8aa15d05b
Web Backend	affaan-m/ecc / skills/api-design	25278e7d7b6c
Web Frontend	anthropics/skills / skills/frontend-design	ca45fc496a34

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INTERPRETATION / REPRODUCTION

What the results establish - and what remains open

The requested >95% AI-quality target has not been established. Some development agents inherited parent chat history that could expose prior analysis or answers. Their rubric percentages are therefore withdrawn from headline comparisons. The raw answers, reviewer decisions and targeted corrections remain preserved as development evidence, not a fair leaderboard.

A replacement balanced comparison was frozen with explicit fresh-context agents (fork_turns=none), the same task/schema and dispatch template for each condition, and no further source tuning. The account usage limit stopped that attempt: 20 of 120 planned answers were captured, all from two baseline batches. There is no complete revised/reference/control arm and no valid comparative score. The pending state and saved inputs are recorded in isolated/STATUS.json and input-manifest.json.

The tasks are known H01-H30 cases, so even a completed replay would be adaptive regression, not new generalization evidence. A fixed served-model ID was not recorded; no repeated-trial uncertainty or external human audit is available. Historical VS Code/Copilot and Antigravity evidence stays separate in web-results/comparison; this cycle did not rerun IDE inference. Earlier natural-language routing remains 18/30. Native, embedded, GPU, deployed-service and full accessibility outcomes remain unmeasured.

Reproduce and inspect

```
python scripts/build_catalog.py --check
python scripts/check_improvement_candidate.py
python -m pytest -q
python scripts/compile_improvement_reviews.py
python scripts/compile_improvement_followup.py
python scripts/report_improvement_cycle.py
```

Current evidence: verified-domain-table.csv; verified-evidence-chart.json; candidate-integrity.json; final-tests.txt; per-run executable/browser logs; distribution-verification.json; isolated/STATUS.json. Earlier review-results.json and final-candidate-results.json are explicitly marked development-only. They must not be cited as independent quality results.

Resume the frozen fresh-context comparison when inference capacity is available; retain any misses. Then use newly authored tasks, repeated model/host runs and blinded independent review. Re-establish 390px rendering and improve the 18/30 natural-language router before broader claims. This is local development validation, not exact-commit remote CI or a published release. GitHub, PyPI and website publication remain held for review and destination approval.

Local artifact fingerprints (full digests in distribution-verification.json): gncore-7.0.0rc1-py3-none-any.whl: ea1dda54e80bea01ab98...; gncore-7.0.0rc1.tar.gz: 2d92081ad4cf377c5ae5...; gncore-7.0.0rc1-skills.zip: 978bdc8c59d91c1622dc...

Completion boundaries - 1 to 8

These are the reviewed instructions added to the domain guides. They describe required engineering behavior; their presence alone is not a measured runtime outcome. Apply only the checks relevant to the requested task.

Domain	Added completion boundaries
AI Engineering	Bind each factual claim to authorized evidence; keep tenant/user scope across retrieval, caches, memory and citations, including access changes. Give tools finite call/time/output budgets and typed empty/error fallbacks; a lost reply to a side effect requires operation identity or reconciliation before retry.
Automation	Scope operation identity to the owner and payload; reject conflicting reuse and make overlapping triggers acquire atomic ownership. Fence stale owners at the effect's storage boundary; retain a bounded retry/quarantine record and verify crash/replay across the durable commit.
Cloud DevOps SRE	Check reader/writer compatibility, least-privilege identity and secret handling before exposure; promote the same tested artifact digest for each intended target. Use workload-specific error, latency, lag and resource gates; distinguish startup/readiness/liveness and preserve a compatible recovery path before contraction. Before delivering a rollout plan, state the credential delivery boundary: a managed identity or authorized runtime secret store, no embedded credentials in artifacts, and redaction from logs. Naming least privilege alone does not cover credential exposure.
Code Build	Match the supplied API, exception semantics, units and mutation behavior; use an independent counterexample for the actual failure mode. For exact allocation/ranking, test ordering as well as conservation, mixed exact/fractional shares, stable ties and invalid inputs; do not infer general correctness from a small fixture.
Computer Vision	Check annotation dimensions, legal labels, empty/degenerate cases and geometric alignment; transform predictions back to the declared original coordinate frame. Keep deployment-relevant groups disjoint, tune only on validation, retain a simple baseline and report task/per-class metrics with sample support.
Cybersecurity	Separate demonstrated exploits, statically established paths and unresolved hypotheses; record attacker control, impact, repair and safe neighboring behavior. Measure false alarms and repaired behavior separately from finding recall; secret removal does not revoke exposed credentials or derived sessions.
Data Engineering	Declare record grain, business time, constraints, revision ordering and tombstones; an older late update must not overwrite newer accepted state. Define watermark/finality and correction/retraction policy; publish idempotently and reconcile raw, duplicate, rejected, accepted and emitted counts/amounts after restart.
Data Science	Reconcile the observational unit, assigned population, denominators and join cardinality before estimating effects; inspect attrition and post-treatment selection. Check quoted arithmetic and uncertainty at the independent unit; choose suitable projected or geodesic calculations for spatial quantities and distinguish inference from causal evidence.

Completion boundaries - 9 to 16

These are the reviewed instructions added to the domain guides. They describe required engineering behavior; their presence alone is not a measured runtime outcome. Apply only the checks relevant to the requested task.

Domain	Added completion boundaries
Databases	Coordinate old/new writers before and throughout resumable backfill, reconciliation and contraction; state the unknown-value policy and last reversible boundary. Verify engine/version lock and constraint semantics; EXPLAIN ANALYZE executes the statement. Retry the whole transaction when required without duplicating external effects.
Deep Learning	Tie model/data/config/environment identity to optimizer, scheduler, RNG, sampler, AMP and accumulation state; use a completed-update restart boundary when partial state is absent. Normalize by actual valid-example/token counts across microbatches/ranks and advance schedules only on completed optimizer updates; compare controlled interrupted and uninterrupted trajectories.
Design UX	Specify narrow-screen content/action order, semantic names, keyboard path, visible/restored focus and persistent linked errors; include consequential secret/destructive states when relevant. Announce async status without surprising focus moves; define focus after dialogs, removals and validation. Separate the proposed interaction specification from tested behavior.
Developer Research	Separate primary version-matched evidence, secondary examples and inference; identify the smallest discriminating experiment, expected observations and stopping bounds. Record assumptions and what would change the recommendation; a failed reproduction is limited to its configuration, and observed behavior does not establish documented support.
Developer Tools	For repository/workspace changes, preflight ownership, dirty paths and collisions; journal only owned partial state and preserve intervening user work during recovery. Use explicit paths and argument arrays; handle option-like values at the invoked program's boundary. Define actionable exit behavior and verify spaces, Unicode and interruption.
Distributed Systems	Write the actual interleaving and transaction boundaries; enforce monotonically increasing ownership generations at the protected write, not only in the lease record. Scope deduplication to operation/payload identity and a replay-compatible retention policy; reconcile unknown outcomes and test expiry, stale owners, overlap and lost replies.
Documentation	Use only the supplied or verified API/help contract; label illustrative schema, status codes and assumptions instead of presenting invented fields as requirements. For procedural/migration guides, include compatible shell/platform examples and a recoverable failure path; preserve scope for narrow edits and distinguish unexecuted examples. Give a self-contained example a reader can copy. If the wire schema is incomplete, clearly label every assumption or supply a runnable offline mock/client example that preserves the known contract; do not imply unverified wire compatibility or leave the only example as non-runnable placeholder prose.
Embedded Systems	Define bounded nonblocking ISR/task ownership and synchronization; volatile alone does not guarantee atomicity or ordering. Make overrun/drop/timeout state observable. State timer width, wrap comparison and observation/interval bounds; distinguish modular elapsed subtraction from signed deadline ordering and host fixtures from measured hardware timing.

Completion boundaries - 17 to 24

These are the reviewed instructions added to the domain guides. They describe required engineering behavior; their presence alone is not a measured runtime outcome. Apply only the checks relevant to the requested task.

Domain	Added completion boundaries
Game Development	Preserve the requested mechanic and movement feel; respect engine units/time integration, input edges and physics ownership without expanding a small fix into a new game. Bound pause/hitch behavior and compare multiple schedules. Render-rate independence alone does not prove deterministic replay, which also depends on input/random/physics state.
Machine Learning	Choose deployment-time and entity boundaries from the prediction moment; fit preprocessing, feature selection, calibration and thresholds only within training/validation scope. Retain an untouched final test and task-appropriate baseline/metrics: classification imbalance and alert budgets when relevant, regression/forecast metrics otherwise; report group support and uncertainty.
MLOps	Promote a compatible immutable model/preprocessor/schema/data/code/runtime bundle and verify missing, reordered or malformed serving inputs against training behavior. Check artifact trust before executable deserialization; hashes establish identity, not safety. Gate on quality and operations, align delayed labels by maturity/cohort and retain a compatible rollback target.
Mobile Development	Preserve versioned drafts/pending state across process death as well as background/resume; cancel or reject stale results across navigation, logout and account changes. Where relevant, use stable virtualized rows and measured optimization; give offline/error/conflict state and durable mutation identity a real recovery path, then test relaunch/retry on the target platform.
Natural Language Processing	Define label units, special/padding masks and whether offsets address original or normalized text; retain mappings through normalization and truncation. Group correlated conversations/documents, preserve label-bearing long content and reconstruct/deduplicate overlapping windows before task-level evaluation; test short, padded and truncated boundaries.
Neural Networks	Match tensor axes, label semantics, dtype and reduction; compute losses in stable logit space rather than taking a log of probabilities that can underflow. For custom derivatives or suspected gradient errors, freeze stochastic/stateful behavior and check central differences away from kinks; reject malformed/nonfinite inputs and verify tiny-batch parameter updates when training changes.
Observability	Propagate or link queue context using the installed API; use bounded metric dimensions, redacted correlation and meaningful latency/error/throughput/queue signals. Define sustained SLO alerts with owner/recovery/runbook and exercise controlled failures. Bound exporter buffers and verify collector outages cannot block requests, exhaust memory or recurse into logging.
Performance Engineering	Verify equivalent outputs and time actual completion, including async/accelerator synchronization where needed; preserve every outcome and recompute quoted statistics from raw samples. Match environment, workload, concurrency, warmup and cold/warm conditions; report relevant tails, throughput and memory against the user's objective, including errors/timeouts and small-sample limitations.

Completion boundaries - 25 to 30

These are the reviewed instructions added to the domain guides. They describe required engineering behavior; their presence alone is not a measured runtime outcome. Apply only the checks relevant to the requested task.

Domain	Added completion boundaries
Python	For ordinary fixes, preserve API/runtime behavior and run relevant regressions without imposing a package release workflow. For packaging/resources/distribution changes, build the wheel from the produced sdist, inspect metadata/resources/entrypoints and clean-install outside the checkout across supported interpreters and working directories.
Release Engineering	For each intended target artifact, link accepted source/version, CI evidence, metadata and hashes; promote those same verified bytes to destinations covered by authorization. Verify clean installs and actual destination availability before updating in-scope claims. Preserve recovery for partial publication; do not overwrite immutable registry files or imply unperformed publication.
Systems Programming	Specify stable storage, ownership, retention, matching allocation/deallocation and destruction after foreign callbacks become quiescent; define permitted concurrency. Distinguish enforceable null/length/alignment/handle/overflow checks from caller lifetime/provenance obligations; specify error/unwind behavior and use appropriate native/sanitizer/stress evidence without claiming pointer inspection proves safety. Write the actual ABI rules rather than saying to document them: whether null is permitted at zero length, required alignment and checked length arithmetic, who owns each allocation, caller lifetime obligations, and a specific failure/status convention that prevents unwinding across the boundary. Mark unspecified foreign guarantees as assumptions.
Testing	Derive observable assertions from an independent contract and meaningful failure transitions; a corrected oracle needs contract evidence, not a desire for green CI. For mutation evaluation, first accept a known-correct implementation, then count caught mutants and false rejections separately. Record actual failures/skips and unavailable integration coverage.
Web Backend	For retryable state-changing work, bind durable keys to authenticated tenant and canonical payload; reject changed reuse and choose one consistent ownership/commit protocol. Commit command/job, outbox and replay outcome at the required atomic boundary; explain duplicate blocking, rollback and lost-commit replies, or lease takeover/fencing for separately committed claims. Scope unrelated backend changes to their own contracts.
Web Frontend	Infer a coherent direction from audience/content/action while preserving explicit brand/stack constraints; implement named controls and actual loading/error/empty/success transitions. Verify selected-value confirmation, keyboard-visible custom controls, a deliberately focusable status target and narrow layouts; stale async responses/optimistic rollbacks must not overwrite newer state. Local feedback must not claim external actions occurred.